

Chapter 5

Nutrition During Infancy



Course objectives

- Discuss the key development milestones of child growth and development.
- Discuss the nutrient needs of new borns, infants and young children.
- Discuss Infant and Young Child Feeding in the context of HIV/ AIDS.
- Discuss the common nutrition-related health issues during childhood.

Introduction

- ❖ The first two years of life provide a **critical window of opportunity for ensuring children's appropriate growth** and development through optimal feeding
- ❖ Malnutrition during the first 2 years of life causes **stunting**
 - ✓ leading to the adult being several centimeters shorter than his or her potential height
- ❖ Adults who were malnourished in early childhood have impaired **intellectual performance**, **reduced capacity for physical work**

Introduction...

Neonate"-- is a newborn

Neonatal period

– Is the period from birth to 28 days

Infancy:0-1 year

Growth: the physical change which occurs as a result of change in number and size of cells in the body

Newborn Nutrition

- ❖ Neonatal Nutrition is important
 - To meet the **metabolic requirements of developing organ systems**
 - Satisfy normal growth requirements and
 - Prevent deficiency
- ❖ The nutritional status of a newborn is determined by intrauterine growth and gestational age.
- ❖ The Nutritional requirements of the **term neonate** differ from those of the **preterm neonate(<37 weeks)**
 - Preterm babies doesn't have enough **nutrient store** so they need **supplement**

Newborn Nutrition

- ❖ Infants who have health problems may need extra energy up to 160–180 cal/kg/day.
- ❖ Neonates need 100–120(cal/kg/d) in order to grow properly

Assessing Newborn Health

Birth Weight as an Outcome

- ❖ The weight of a newborn is a key measure of health status during pregnancy.
- ❖ Full-term infants usually weigh 2500–3800 grams and are 47– 54 centimeters in length.
- ❖ The greatest risk to life is in its beginning (before, during and in the very first hours and days after birth).



- ❖ In institutional settings, **preterm newborns** are kept in the neonatal intensive care unit (NICU), and get:
 - ✓ temperature,
 - ✓ the right balance of fluids,
 - ✓ minerals (electrolytes), such as sodium and potassium until their bodies mature and they are able to suckle breast milk

The key Development Milestones

- ❖ Growth and development involve a series of changes by which the fetus becomes a mature organism.
- ❖ Normal growth and development is characterised by a
 - ✓ regulated increase in the size, mass and
 - ✓ complexity of function of tissues and organs

Fetal Growth From 4 to 40 Weeks



The key Development Milestones...

- ❖ Growth and development is absolutely dependant on a supply of **energy and nutrients**.
- ❖ Any limitation of **nutrient supply** and **disease** can constrain the pace and pattern of development.
- ❖ An infant's birth weight doubles in the first 4 to 6 months and triples by the end of the first year.
- ❖ During the first 4 to 6 months, almost one third of the infant's energy requirement is used for growth.

Infant Feeding

- Energy (Calories):
 - 108 kcal/kg/day from birth to 6 months and 98 kcal/kg/day from 6 to 12 months
- Factors that influence calorie needs.
 - Growth rate
 - Sleep/wake
 - Temperature and climate
 - Physical activity
 - Health status

Nutrient requirement

❖ Protein Needs

2.2 g/kg/day from birth to 6 months

1.6 g/kg/day from 6 to 12 months

❖ How much is that?

✓ Newborn weighing 4 kg needs $2.2 \times 4 = 8.8$ g protein

✓ 6-month-old weighing 8 kg needs $1.6 \times 8 = 12.8$ g protein

❖ And human milk provides all the amino acids essential for proper growth in the first 6 months of life.

Nutrient requirement

Infants require high quality protein :

- Build, maintain, and repair new tissues, including tissues of the skin, eyes, muscles, heart, lungs, brain, and other organs.
- Manufacture important **enzymes, hormones antibodies**, and other components and Perform very specialized functions in regulating body processes.

Fat

- ❖ **30 g/day** of fat
- ❖ Requirements for fatty acids on a per-kilogram basis are higher in infants than adults

Fat restriction is not recommended.

Breast milk provides 50% of its calories from fat, and this percentage reflects an adequate intake of fat by infant.

Infants require lipids in their diets because they:

- ✓ Supply a major source of energy –
- ✓ Allow for the absorption of the fat-soluble vitamins A, D, E, and K; and
- ✓ Provide essential fatty acids
 - Breast milk contains short-chain and medium-chain fatty acids (in addition to the long-chain)
 - Easier to digest and utilize than long-chain fatty acids

Nutrient requirement

Vitamins and minerals

- ❖ The vitamin and mineral needs of children **increase with age .**
- ❖ A balanced diet of nutritious foods can meet children's needs for these nutrients
- ❖ With the exception of vitamin D, the vitamins in breast milk are ample to support infant growth

Nutrient requirement

Vitamins A

Vitamin A is essential for:

- ✓ Formation and maintenance of healthy skin, hair, and mucous membranes
 - ✓ Proper vision
 - ✓ Growth and development and
 - ✓ Healthy immune and reproductive systems
-
- Throughout infancy **350 mg** per day is recommended.

Nutrient requirement^t

┐ Vitamin D

- ✓ It is required for bone mineralization with calcium.
- ✓ The highest risk for vitamin D deficiency are those with **darker skin and no exposure to sunlight.**
- ✓ Recommended level of 400 IU per day.

❖ **Vitamin D–deficiency rickets** can cause significant morbidity, including delays in growth and motor development, failure to thrive, short stature, seizures, and skeletal deformities.

Nutrient requirement

❖ Vitamin K

- ✓ injection given at birth to prevent Haemorrhagic disease of the Newborn(HDN), since the infant intestine is sterile ,it is free from bacteria that produce vitamin k .

❖ Calcium

- The calcium content of breast milk is ideal for infant bone growth, and the calcium is well absorbed.

Nutrient requirement

❖ Iron

- ✓ Needed for growth, expansion of blood volume and for improving iron stores in the body
- ✓ Breastmilk contains relatively small amounts of iron, but the iron has a **high bioavailability**
- ✓ Infants born with good iron stores sufficient to last for 4-6 months.
- ✓ 1mg / kg body weight/day.

❖ Zinc also has a high bioavailability, due to the presence of a **zinc-binding protein**

B complex vitamins and vitamin C

- ❖ Breast fed infants derive the required B complex vitamin requirements from breast milk.

Thiamin (vitamin B1), is needed by infants to:

- ✓ Help the body release energy from *carbohydrates during metabolism* and
- ✓ Play a vital role in the normal functioning of the *nervous system*.

Riboflavin (Vitamin B2)

- ✓ Riboflavin (vitamin B2), a water-soluble vitamin, helps the body release energy from **protein, fat, and carbohydrates** during metabolism.

Vitamin C

- ✓ 25mg of vitamin C per day
- ✓ Vitamin C have effect on non-hemiron absorption

Optimal Infant and Young Child Feeding (IYCF)

- ❖ IYCF, refers to nutrition and care techniques that improve child survival and growth.
- ❖ Adequate nutrition during infancy and early childhood is essential to ensure the growth, health, and development of children to their full potential.
- ❖ It is estimated that sub-optimal breast-feeding, especially **non-exclusive breastfeeding in the first 6 months of life**, results in
 - ✓ 1.4 million deaths and 10% of the disease burden in children younger than 5 years

BREAST FEEDING

- Mothers should exclusively breastfeed their infants for the **first 6 months** of life and thereafter, continue breastfeeding up to **24 months or beyond**.
- Exclusively breastfeeding- giving no other food or drink – not even water – except breast milk(WHO).
 - Drops or syrups consisting of vitamins, mineral supplements or medicines are permitted

- ❖ Human milk contains active cells, anti-infective factors, a large number of hormones and growth factors and at least 60 enzymes.
- ❖ Breast milk from a woman who is in good health and nutritional status provides a complete food for healthy infants during the early months of life.

Benefits of breastfeeding

FOR THE BABY:

- Improved growth and nutrition status
- Less likely to die
- Increased bonding
- Lower risk of chronic diseases (diabetes, heart disease, asthma, some cancers)
- Less diarrhoea and respiratory infections
- Lower risk of overweight/obesity
- Less ear infections, GI disorders, skin conditions and SIDS
- Improved cognitive and motor development



FOR THE MOTHER:

- Mother less likely to become pregnant in early months
- Faster maternal recovery and weight loss post partum
- Lower risk of maternal cancers (ovarian and breast cancer)
- Less post-partum depression

Optimum breast feeding

1. Initiate breastfeeding immediately
2. Give colostrums
3. Practice exclusive breastfeeding from 0-6 months
4. **Practice frequent, on-demand breastfeeding**
5. Good attachment *and positioning*

Formula-feeding

Condition to use breast milk substitute

- ❖ Some women may not be able to breast-feed
 - Women with AIDS, human T-cell leukemia, or active tuberculosis, who are receiving chemotherapy and/or radiation, or who use illegal drugs should not breast-feed
 - Infants with galactosemia cannot metabolize galactose and lactose and should not be breast-fed
 - Soy-based formulas is recommended
- ❖ Formula can be an alternative to breast-feeding
 - Developed to be as similar as possible to breast milk

Formula-feeding

- ❖ Infant formulas are manufactured to closely **resemble human milk**; however, numerous components cannot be replicated
- ❖ Formula can be an alternative to breast-feeding. Use of **iron-fortified formulas**, not low iron formulas is recommended.
- ❖ Special formulas are also available for **low birth weight infants** and **infants with specific inborn errors of metabolism**.

Formula-feeding

- ❖ **Soy-based formulas** made from soy protein, vegetable oils and glucose polymers ($\pm$ sucrose).
- ❖ They are designed to be use by infants who are **allergic to cow's milk and infants** in vegetarian families, infants who suffer from **galactosemia or lactose intolerance**.

Cow's Milk

- ❖ Cow's milk is not recommended for infants less than 1 year of age due to:
 - ✓ high protein content accompanying **renal solute load**;
 - ✓ the risk of an **allergic reaction**; and
 - ✓ the risk of a **blood-losing enteropathy** in some infants.
- ❖ Skim and partially-skimmed cow's milk are not suitable for infants less than 2 years of age as they are of **low energy density, lack the essential fatty acids and have high renal solute load**.

Guidelines for feeding infants safely

- ✓ Infants under 1 year of age should not be fed **honey** to prevent them from botulism
- ✓ **Unpasteurized milk** or food should not be fed to infants as they can introduce pathogens (*E. coli 0157:H7* & *Salmonella*) cause diarrhoea or other more serious infections.
- ✓ Infants should not be fed **raw eggs** to prevent them from salmonella poisoning.
- ✓ Avoid bottle feeding & supervise infants during feeding.

Complimentary feeding

- ❖ CF---is giving a food in addition to Breast milk.
- ❖ **At 6 months of age:**
 - Breast milk alone does not provide sufficient calories & nutrients to sustain optimal growth;
 - CF becomes necessary to fill the energy and nutrient gap
 - Infants are developmentally ready to take soft, and semi-soft foods in addition to breast milk.
- ❖ In many countries, the period of complementary feeding from 6–23 months is the time of **peak incidence of growth faltering, micronutrient deficiencies and infectious illnesses**

- Introduction of solid and semi-solid foods begins at approximately six months of age
 - Infant cereal mixed with breast milk, formula, or water is common
 - Single-ingredient foods are recommended

Complimentary feeding...

- ❖ Guiding principles for **CF...reading assignment**
- ❖ Complementary foods should be:
 - ✓ **Timely**
 - ✓ **Adequate**
 - ✓ **Safe**
 - ✓ **Properly fed**

CHILD FEEDING AND HIV/AIDS

- ✓ Energy requirements in children can vary according to the **type and duration of HIV**-related infections.
- ✓ Similar to asymptomatic HIV-infected adults an average increase of **10% of energy** intake is recommended for infants to maintain growth.
- ✓ To achieve **catch-up growth** in children, energy intakes for HIV infected children experiencing weight loss need to be increased by **50% to 100%**

CHILD FEEDING AND HIV/AIDS

- Replacement feeding prevents BM transmission of HIV,
 - But in resource limited settings, the risk of death from artificial feeding must be weighed against the risk of HIV infection.

CHILD FEEDING AND HIV/AIDS

- When replacement feeding is **AFASS** avoidance of breastfeeding by HIV-infected mothers is recommended.

Acceptable:

The mother perceives no problem in replacement feeding.(cultural, social, or due to fear of stigma and discrimination.)

- **Feasible:**

The mother (or family) has adequate **time, knowledge, skills, resources and support** to correctly mix formula or milk and feed the infant up to 12 times in 24 hours

- **Affordable:** The mother and family can pay the cost of replacement feeding without harming the health or nutrition status of the family.
- **Sustainable:** Availability of a continuous supply of all ingredients needed for safe replacement feeding for up to one year of age or longer.
- **Safe:** Replacement foods are correctly and hygienically prepared and stored, and fed preferably by cup.

Infant feeding options from 0 to 6 months

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graph TD; A[Infant feeding options from 0 to 6 months] --> B[Replacement feeding when AFASS:]; A --> C[Exclusive breastfeeding]; A --> D[Other breast milk options:]; C -.-> E[Early cessation when RF becomes AFASS];
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Replacement feeding when AFASS:

- Commercial infant formula
- Home-modified animal milk with a micronutrient supplement

Exclusive breastfeeding

Early cessation when RF becomes AFASS

Other breast milk options:

- Expression and heat-treatment
- Wet-nursing (breastfeeding by an HIV-negative woman)
- [Milk banks]

Common Nutritional Issues and Concerns

❖ Iron deficiency and iron-deficiency anemia

- ✓ Inadequate iron contributes to long-term learning delays
 - Role in central nervous system development
 - Associated with poor cognitive, motor and socio-emotional development, as well as
 - poor cognition and school achievement

❖ Diarrhea and constipation

- ✓ Diarrhea: 3 or more loose, watery stools per day
 - May result infections, food intolerance, changes in fluid intake, or excessive fruit juice intake
- ✓ Constipation: changes in the frequency, size, consistency, or easy of passing stool

❖ **Food allergies**

- ✓ About six to eight percent of children under four years old have allergies
- ✓ Absorption of intact proteins is the basis
- ✓ Common symptoms: wheezing or skin rashes
- ✓ Guidelines recommend breast milk for infants considered at risk

❖ **Lactose intolerance**

- ✓ Inability to digest lactose
- ✓ Characterized by cramps, nausea, pain, and alternating diarrhea
- ✓ and constipation
- ✓ Uncommon in infancy and tends to be overestimated



Chapter 6

Nutrition of Pre-school Children

Definition

- **Toddlers** : child age 1-3 years
- **Preschool Child**: child age 3 – 6 Years

Growth pattern

- ✓ The rapid growth during infancy is followed by a generally slow growth between one to six years.
- ✓ The child becomes more active and the social and environmental influences have a great impact on his food behavior and eating pattern.
- ✓ The need for nutrients is increased as growth and development continues.
- ✓ During the second year, the increase in height is about 10cm and weight gain is 2 to 2.5 kg.
- ✓ After two years annual gain in height and weight is only six to seven cm and 1.5 to 2 kg respectively.

Growth pattern

- ✓ As growth proceeds, changes occur in proportion of water, muscle tissue, fat deposits and skeletal structure.
- ✓ The body water gradually decreases and there is addition of **adipose tissue** and that of **minerals to the bones**.

Preschool nutrition

- ✓ Preschool age is also a period of imitation and sex identification, with boys imitating their fathers and girls their mothers
- ✓ Such behaviour is reflected at meal times, therefore the parents play an important role inculcating healthy and positive attitude towards food

Nutritional Requirements in Preschool Children

After 1 yr of age,

- The child's rate of growth slows and the nutrient needs for growth consequently decrease.
- The rate of growth remains appreciable.
- Activity increases.

Nutritional requirements of preschool children

Energy

- **Energy is required for growth and activity.**
- Individual children's energy needs vary widely, depending on their growth and physical activity.
- **The energy requirements are**
 - ✓ **1-3 years = 102 k.cal/kg body weight or 1300k.cal for and**
 - ✓ **4-6 years 90 k.cal/ kg or 1800 k.cal /day**
- Total energy needs increase gradually with age, but energy needs per kilogram of body weight actually decline.

Energy

(EER) for children ages 13–36 months is

- $(89 \times \text{weight of child [kg]} - 100) + 20$ (kcal for energy deposition).
- For example, a healthy 24- month-old girl who weighs 12 kg would have an EER of $(89 \times 12 \text{ kg} - 100) + 20 = 988$ kilocalories.

NUTRIENT NEEDS CONT...

- Physically active children of any age need more energy because they expend more and inactive children can become obese even when they eat less food than the average.
- Some children, notably those adhering to a **vegan diet**, may have difficulty in meeting their energy needs.
- **Grains, vegetables and fruits** provide plenty of fiber, adding bulk, but may provide too few kilocalories to support growth.
- **Soy products, other legumes and nut or seed butters** offer more concentrated sources of energy to support optimal growth and development.

Protein

- The increasing muscle mass requires a positive nitrogen balance.
- The additional requirement for growth have been considered in determining the allowances for protein.
- Dietary Reference Intakes for protein
 - 1–3 years 16 g/day
 - 4–8 years 24 g/day

Fat

- ❖ Fat in a young child's diet can be achieved by employing the principles of **Food Guide Pyramid** that promote a diet of
 - whole-grain breads and cereals, beans and peas,
 - fruits and vegetables,
 - low-fat dairy products after 2 years of age, and lean meats
- ❖ Adequate fat is required to provide the extra calories as well as to **add calorie density** and reduce bulk

Fiber

- ❖ fiber in a child's diet helps prevent constipation and is part of a healthy diet.
- ❖ Too much fiber in a child's diet can be detrimental because
 - have the potential of reducing the **energy density of the diet**, which could impact growth.
 - impact the bioavailability of some minerals, such as **iron and calcium**
- ❖ 1–3 years of age 19 g/day of total fiber
- ❖ 4–8 years of age 25 g/day of total fiber

Calcium

- ❖ Adequate calcium intake in childhood affects peak bone mass.
- ❖ A high peak bone mass is thought to be protective against osteoporosis and fractures later in life.
- However, many children do not consume adequate calcium children 2-8 yrs.
 - soft drink and juice consumption begin to increase at 4 to 8 years of age, often replacing milk at meals and snacks.
 - ✓ 500 mg/day for children ages 1 to 3 years.
 - ✓ 800mg/day for 4-8 years.

Iron

- ✓ Daily requirements for iron intake are based on age and iron stores.
- ✓ During periods of rapid growth the body's need for iron increases,
- ✓ Adequate iron intake is important in this age group to prevent iron deficiency.
- ✓ Iron requirements during childhood need to cover the amount needed for growth and for increase in the hemoglobin concentration by about 1g/ dl.

Table 10.6 Dietary Reference Intakes for key nutrients for toddlers and preschoolers^{32,33}

Age	Recommended Dietary Allowances		Adequate Intake
	Iron (mg/d)	Zinc (mg/d)	Calcium (mg/d)
1–3 years	7	3	500
4–8 years	10	5	800

Vitamins

- **Vitamin A**----400g of vitamin A/day.

Vitamin D

- is available to humans through the photochemical action of sunlight on **7-dehydrocholesterol** in skin and
- through dietary sources such as fish oils, fatty fish, and foods fortified with vitamin D, including cow's milk and infant or supplemental formulas
- Vit D is important for absorption of calcium

B complex vitamins

- ❖ The recommended intake for B vitamins is based on the energy intake and is calculated as for adults
- ❖ i.e , 0.5mg/1000k.cal, 0.6mg/1000k.cal and 6.6mg/1000k.cal for thiamine, riboflavin and niacin respectively.
- ❖ Folic acid and pyridoxine requirement for children is slightly higher than that of infants.

➤ Fluids

- Healthy toddlers and preschoolers will consume enough fluid through beverages, foods, and glasses of water to meet their needs.
- Fluid requirements increase with fever, **vomiting, diarrhea, and when children are in hot, or dry environments**
- They prefer to take **carbonated drink than milk** at this age group

Healthy foods

- Preschoolers can eat what the rest of the family eats
- A young child's eating plan should consist mostly of healthy foods, such as lean meats, poultry, seafood, eggs, and legumes; whole grains, such as whole-wheat bread and cereals; at least two servings of dairy foods daily; and fresh or lightly processed fruits and vegetables

feeding pattern

- ❖ Children learn by observing the behaviors of others and then incorporating and imitating these practices.
- ❖ Parental influence in showing dislike for a particular food
- ❖ Food choices and dietary quality in the early years of life are usually influenced by parents and environment
- ❖ Parents modeling healthy behaviors, accessibility to sedentary pursuits at home, sibling influences, and family television viewing habits may be important determinants of a child's activity level and healthy behaviors .

Role Models CONT....

- Modeling food behaviors is one of the most effective ways to promote increased consumption of healthy foods .
- Parents who do not eat fruits and vegetables tend to raise children who refuse these foods as well.

➤ Food Refusal

- Commonest nutrition complaint at this age.
- Preschool children may show certain likes and dislikes which when not properly interpreted may lead to **under nutrition**.



➤ Food fads

- is when your **child** is favours one **food** over everything else over a length of time and refuses to eat anything else
- These food fads are one of the ways children try to assert themselves as individuals.

What to do:

- **Make food fun**

For a toddler, enjoying food means touching, feeling and playing with it. Let children feed themselves.



- **Set an example**

Toddlers are very good imitators

- **Avoid too many drinks and snacks**
- *Offer meals at regular meal and snack times. .*
- *Just getting children to taste the food will eventually lead them to accepting most foods in the future.*
- **Colorful foods add variety and stimulate interest**



Malnutrition

- Child malnutrition is the single biggest contributor to under-five mortality due to greater susceptibility to infections and slow recovery from illness.
- They do not reach their optimum size as adults (and so may have less physical capacity for work)
 - Child malnutrition impacts on education attainment.
 - The degree of cognitive impairments is directly related to the severity of stunting and Iron Deficiency Anaemia.
 - lower cognitive test scores, delayed enrolment,
 - higher absenteeism and more class repetition compared with non stunted children.

- Vitamin A deficiency reduces immunity and increases the incidence and gravity of infectious diseases
- Child malnutrition impacts on economic productivity.
- The mental impairment caused by iodine deficiency is permanent and directly linked to productivity loss.
- ✓ The loss from stunting is calculated as 1.38% reduced productivity for every 1% decrease in height
- ✓ while 1% reduced productivity is estimated for every 1% drop in iron status (source Haddad and Bouis, 1990).

- ❖ Protein energy malnutrition, micronutrient deficiencies such as **vitamin A deficiency** and iron **deficiency anemia** are common among preschool children
- ❖ The primary cause of malnutrition is inadequate diet and another factors include:-
 - ✓ Chronic infection
 - ✓ Poor environmental sanitation
 - ✓ Poor personal hygiene

PEM

- range of pathological conditions arising from coincident lac of varying proportions of protein and calorie

🕒 PEM is protein and calorie DEFICIENCY, occurring most frequently in infants and young children and often associated with infection.

🕒 The symptoms vary with the degree of malnutrition. Symptoms of different types of PEM are as follows.

1. Kwashiorkor:

Oedema of face and lower limbs, failure to thrive, anorexia, diarrhea, dermatitis, apathy, sparse, soft and thin hair, angular stomatitis, cheilosis and anemia.

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Protein Energy Malnutrition

2. Marasmus:

- Failure to thrive, irritability and apathy, diarrhea, dehydration.
- The child is shrunk and there is little or no subcutaneous fat.

3. Marasmic kwashiorkor:

- These children exhibit symptoms of both marasmus and kwashiorkor.
- The peak prevalence of protein energy malnutrition is observed in the age group of 2-3 years and marasmus between 1-2 years

Vitamin A deficiency

- ⌚ Inadequate dietary intake of vitamin A or its precursor (b -carotene) is the most contributory factor for vitamin A deficiency.
- ⌚ Blindness due to xerophthalmia is an important public health problem.
- ⌚ The term xerophthalmia includes all ocular manifestations of vitamin A deficiency like night blindness, xerosis, and Bitot's spot and corneal xerosis.
- ⌚ Vitamin A deficiency increases the risk of disease and death.
- ⌚ This deficiency lowers children's resistance to common infections and results in increased levels of child mortality

Iron deficiency anaemia

- Iron is important in the formation of healthy red blood cells.
- It combines with protein to form haemoglobin found in the blood .
- adequate supply of iron help to our body to have resistant to infection
- Iron- deficiency anaemia is the most common nutritional deficiency in children
- A common cause of iron- deficiency anaemia in the 1 to 2 year old is due to **excessive milk intake**.
- *From 2-3 years due to lack of iron- rich foods in the diet*

Causes of anemia

Most common cause of anemia

- ❖ *Inadequate dietary intake*
- ❖ *Poor absorption*
 - *Low iron bioavailability*
 - » *Non-heme iron*
 - » *Inhibitors*
- ❖ *Blood losses(Malaria, parasite and menses)*

IDA...

- The symptoms include
- Skin appears pale
 - ✓ Fingernails become thin and flat and eventually results in koilonychias (spoon – shaped nails)
- Long term deficiency symptoms result in malfunctioning of various systems.
- The general symptoms are lassitude, fatigue, and breathlessness on exertion, palpitation, dizziness, and headache, dimness of vision, insomnia and angina.

Health consequence Of IDA

- ✓ poor growth
- ✓ Poor resistance to infection
- ✓ poor productivity
- ✓ Poor cognitive development.
- ✓ Impair academic achievement
- ✓ IDA lowers IQ by 10 points
- ✓ Increase absentees from school

Strategies

1. Supplementation of iron and folic acid
 2. Treatment of severe anemia
 3. Dietary diversification
 4. Fortification of foods with iron
 5. Control of malaria (prophylaxis and treatment), helminthiasis, and schistosomiasis.
- Sustainable strategies for preventing IDA **Food based approaches** to increase iron intake through
 - Food fortification and
 - Dietary diversification in the general population.

Test(5%)

1. write the cause of Iron Deficiency anemia in preschool children
2. why too much fiber is not recommended for preschool age children
3. write the impact of child malnutrition?



*“Train a child in the way he should go,
and when he is old, he will not depart
from it.”*

Chapter seven

NUTRITION DURING SCHOOL Age

At the end of this chapter you will be able to:

- Explain the physical and social developments occurring during school-aged children.
- Describe the role nutrition in the physiological and psychosocial development of school age children.
- Relate nutritional needs of school-aged children and their patterns of food intake.
- Explain the recommended energy and nutrient needs of school-age children.
- Identify nutrition and health problems of school-aged children and their prevention.

Introduction

Age 6 to 12 years

- ✓ Growth is slow and steady.
- ✓ The average annual growth (3–3.5 kg) in weight and 6 cm in height.
- ✓ Children begin to make their own food choices
- ✓ Activity levels vary

GROWTH & DEVELOPMENT

- ❖ The slowed rate of growth during this period results in a gradual decline in the food requirement per unit of body weight.
- ❖ Reserves are laid down for rapid growth during next phase.
- ❖ As child steps in SCHOOL AGE PERIOD, he develops :
 - Ability to work out problems
 - Team work ability

Physical and Social Development

1. Physical growth and development

- ❖ covers the aspects of development in terms of size, height and weight.
- ❖ With growth, there is also a dramatic change in body proportions.
- ❖ Physical development also covers the important skills of using our bodies; the use of large muscles to walk, run, climb, jump and use of our smaller muscles, for example for hand co-ordination.

Physical growth and development

- ❖ During middle childhood, muscular strength, motor coordination, and stamina increase progressively.
- ❖ During the early childhood years, percent body fat reaches a minimum of 16% in females and 13% in males.
- ❖ Percent body fat then increases in preparation for the adolescent growth spurt.
- ❖ The increase in percent body fat with puberty is earlier and greater in females
- ❖ At age of 11-12 year girls are usually heavier and taller than boys.
- ❖ During middle childhood, boys have more lean body mass per centimeter of height than girls do.

2. Social and Emotional Development

Socialisation

- **Primary socialisation** takes place within the family, in the first years of a child's life.
 - ✓ This helps children to learn how to interact with others, what is acceptable and what is not.
- **Secondary socialisation** starts when children come into regular contact with people and settings outside their home like playgroup, nursery and school, and continues throughout life.
 - ✓ Children observe other children and adults' behaviour, and develop a sense of right and wrong.

Energy and Nutrient Needs of School-Age Children

- ❖ School-aged children continue to need healthy foods and nutritious snacks.
- ❖ Now is time to relax and get ready for the teen years, but is also the time to make sure your child is getting great nutrition to prepare for the teen years

Energy and Nutrient Needs of School-Age Children

Energy Needs

- ❖ Energy need of an individual child is dependent on the child's gender, age, height, weight, activity level and body size.
 - Eg, an 8-year-old girl; weight (25.6 kg); 128 cm tall requires
 - 1360 kcal/day if sedentary,
 - 1593 kcal/day if she is low-active,
 - 1810 kcal/day if active, and
 - 2173 kcal/day if very active.

Energy Needs...

- ❖ Energy allowances based on body weight are lower for school-age children than for toddlers and preschoolers.
 - ✓ This reflects the slowing growth rate of school aged children.
- ❖ Younger school age children (7-9 years) require more energy per kilogram of body weight (67 Kcal) compared to 10-12 years old (60 Kcal).

GENDER	AGEGROUP	Kcal/day
Children (boys and girls)	7-9 year	1690
BOYS	10-12 year	2190
GIRLS	10-12 year	2010

Fat

- ❖ Total fat requirement is below 25 % of total energy is considered to affect growth in children.
- ❖ To provide 25 % total fat calories, a minimum level of visible fat in children should range between 25-35 g/day
- ❖ Reduce intake of saturated and trans fats and increase unsaturated (MUFA & PUFA).

Proteins

- ✓ The recommended protein intake in children 6 to 10 years is $0.95 \text{ gm/kg/day} = 28\text{g/day}$.
- ✓ Girls require more protein between 10-12 years than boys for approaching menarche.

Vitamins and Minerals

- According to food-consumption surveys, children's mean intakes of most nutrients meet or exceed the recommendations.
- But still, certain subsets of children do not meet their needs for key nutrients such **as iron and zinc**, which are important for growth, and **calcium**, needed to achieve **peak bone mass**.

Recommendations for intake of Iron, Fiber, Calcium, Vitamin D and Fluids

Iron

- Although iron deficiency is not as prevalent during the school-age years as it was during the toddler and preschool-age years, adequate intake of iron is still important.
- Consumption of iron-rich foods; such as meats, fortified breakfast cereals, dry beans and peas, vitamin C rich foods are recommended.

Calcium and Vitamin D Needs of School-aged children

- ❖ Adequate calcium intake during this time is necessary to achieve peak bone mass, which may prevent **osteoporosis** later in life.
- ❖ The recommendations for adequate daily intakes of calcium are
 - ✓ 800 mg for children aged 4 to 8 years and
 - ✓ 1300 mg for children 9 through 18 years.
 - ✓ **2-3 glasses of milk** should be taken per day
- ❖ The higher recommendation for older children reflects the fact that most **bone formation** occurs **during puberty**

Table 12.2 Dietary Reference Intakes for key nutrients for school-age children^{28,29}

Age	Recommended Daily Allowances		Adequate Intake
	Iron (mg/d)	Zinc (mg/d)	Calcium (mg/d)
4–8 years	10	5	800
9–13 years	8	8	1300

- ❖ Fiber prevents the occurrence of chronic disease (heart disease, certain cancers, diabetes & hypertension) in adulthood.
- ✓ High-fiber fruits, such as apples with peels; 3g fiber per serving.
- ✓ Vegetables, such as broccoli and whole-grain breads, cereals & brown rice have about 2.5 grams per serving.
- ✓ Dried beans and peas have 4 to 7 grams of fiber ½ serving.

Eating Patterns of School-age children

- Once children enter school, their eating patterns become more scheduled, and the consumption of regular meals especially breakfast becomes an important focus.
- A fortified ready-to-eat breakfast cereal is typically the greatest source of iron, vitamin A, and folic acid for children.
- Nutrition education begins in the home as parents/ other caregivers provide a healthy, well-balanced diet.

Eating Patterns of School-age children

- ❖ When children are young, their parents and families have greater control over what they eat.
- ❖ As they get older, however, what their friends eat in the school environment, and what is available to them in school, will have an impact on what they eat.
- ❖ School-aged children consume more than a third of their calories from saturated fats, one-fourth of the vegetables that children consume are french-fried potatoes.

❖ Skipping breakfast affects their performance level and the calorie and nutrients lost cannot be made up at any other time during the day.

❖ Children who eat breakfast are more likely to meet their needs for vitamins and minerals compared to children not eating breakfast.

Snacking

- ❖ Snacking is becoming the main eating style among children.
- ❖ School-age children consume snacks primarily after school.
- ❖ As the number of snacking increased, the total energy consumption for the children increased.
- ❖ Snacking can be part of a healthy diet if snacks are chosen correctly.
- ❖ Snacks continue to contribute significantly to a child's daily intake.

Prevention of Nutrition and Health Problems

1. Overweight and Obesity

- Significant weight gain generally begins between ages 5 and 7, during puberty or the teenage years.
- Childhood obesity is a serious health threat, because about 40% of obese children become obese adults.
- Besides social difficulties overweight can cause physical problems for children.

- ❖ One in five overweight children children already has health problems related to their weight.
- ❖ Significant health problems are associated with obesity
- ❖ which usually will appear in adulthood, such as:
 - ✓ Cardiovascular disease,
 - ✓ Type 2 diabetes,
 - ✓ Hypertension,
 - ✓ Difficulty moving, and
 - ✓ Trouble sleeping & breathing.

Dental caries

Causes of Dental caries

❖ Bacteria: Streptococcus mutans.

- ✓ Children are often exposed to the bacteria during their infancy.
- ✓ The bacteria in the mouth then break down dietary CHO
 - producing acid that attacks the tooth & can remove minerals from the tooth causing the enamel to weaken and decay.

❖ Chewing candies/ caramels

❖ Lack of fluoride

❖ Poor teeth Hygiene

- Proper cleaning of teeth is important preventive method

Diabetes

- Physicians have noted an alarming increase in the frequency of type 2 diabetes among children (and teenagers).
 - due to the rise in obesity (85%) in this age group & sedentary lifestyle).
- ✓ Appropriate diet and lifestyle intervention should be implemented, along with the use of diabetes medications when necessary.
- ✓ A regular intake of low-glycemic-load fruits, vegetables, whole-grain breads and cereals is especially recommended.

Lactose Intolerance

- ❖ Lactose intolerance is Lactose malabsorption

- ❖ commonly seen in older children than in younger Children.

- ✓ It is a clinical syndrome of gastro intestinal symptoms, such as abdominal pain, diarrhea, nausea, flatulence or bloating.

- ❖ **Primary lactase deficiency:** caused by lack of lactase enzyme.

- ✓ Common in certain racial/ ethnic groups like Jewish people (60 to 80%), Asians & Native Americans (nearly 100%).

- ❖ **Secondary lactase deficiency:** caused by injury to the small bowel, due to an acute infection.

Yogurts, cheeses, and lactose-reduced dairy products are well *tolerated*.

key point

- School-age and preadolescent children continue to grow at a slow, steady rate until the adolescent growth spurt.
- The prevalence of overweight and obesity continue to increase in the school-age and preadolescent groups.
- Monitoring BMI-for-age percentiles is important for screening for overweight or underweight.
- A child's food choices are being influenced by peers, teachers, coaches, the media, and the Internet.
- School-age and preadolescent children are encouraged to eat a variety of foods and increase physical activity